

Progress tracker — Analysis of Algorithms

Status key: ☐ not started ·  in progress · ☒ done · ★ expansion opened

Module	Chapter	Notes	Problems	Status
0	Foundations toolbox	☒ <code>module-00-foundations.md</code>	☒ <code>module-00-problems.md</code>	 start here
1	1 · Analysis of Algorithms	☒ <code>module-01-analysis-of-algorithms.md</code>	☒ <code>module-01-problems.md</code>	☐ after Module 0
2	2 · Recurrence Relations	☐	☐	☐
3	3 · Generating Functions	☐	☐	☐
4	4 · Asymptotic Approximations	☐	☐	☐
5	5 · Analytic Combinatorics	☐	☐	☐
6	6 · Trees	☐	☐	☐
7	7 · Permutations	☐	☐	☐
8	8 · Strings & Tries	☐	☐	☐
9	9 · Words & Mappings	☐	☐	☐

Book exercises (reference)

All 544 end-of-section exercises are extracted per chapter under `exercises/` (`chapter-1-...md` ... `chapter-9-...md`). Use for self-testing alongside each module.

Expansions (sub-chapters)

Opened on demand — none yet.

Log

- **2026-07-11** — Course set up. Stance set to self-contained/thorough. Foundations (Module 0) + Module 1 notes & problems ready. 544 book exercises extracted per chapter. Reference shelf stocked (CLRS, TAOCP 1-4A, Algorithms 4e, Analytic Combinatorics, Zingaro). Starting point: Module 0.

Session log (30-45 min blocks)

Target: 3 sessions/week → ~6 months (started 2026-07-11; projected finish $\approx$ Jan 2027). Tick each as you finish. ~ 77 sessions total (see SYLLABUS "Pace & sessions"). Each session follows `SESSION-PROTOCOL.md` (frame → read → recap → exercises). Module 0 & 1 are broken out concretely below; later modules get split when we reach them.

Module 0 — Foundations (~3)

- ☐ 0.1 — Notes §0.1 Sums + §0.2 Counting
- ☐ 0.2 — Notes §0.3 Binomial coefficients + §0.4 Probability & expectation
- ☐ 0.3 — Notes §0.5 Harmonic + §0.6 Logs/Big-O, then problems P1-P10

Module 1 — Analysis of Algorithms (~4)

- ☐ 1.1 — Notes §1-2 (big idea + cost model)
- ☐ 1.2 — Notes §3 (quicksort recurrence: set up *and* solve, by hand)
- ☐ 1.3 — Notes §4-6 (harmonic numbers, caveats, takeaways) + problems P1-P2
- ☐ 1.4 — Problems P3-P7 (reproduce the quicksort solve unaided)

Spaced-review queue

Every **third** session opens with a 5-min cold re-derivation of the oldest item here (see `SESSION-PROTOCOL.md`). Add a line when a key result lands; move it to the bottom with a new date after each review. Activates $\sim$ session 1.3.

Result / skill	Learned	Last reviewed
<i>(fills in as we complete sessions — e.g. "solve a first-order recurrence", "quicksort compare recurrence")</i>	—	—

Where I am right now

→ **Session 0.1** (in progress — running live with Claude). Format: frame → read §0.1–§0.2 → recap (notes closed) → two exercises. No spaced-review pull yet (nothing prior to recall).